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Effect of Network Topology

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Introduction

Population-level pairwise models characterise a network solely through summary statistics — the mean degree $\langle k \rangle$, the second moment $\langle k^2 \rangle$, and possibly the clustering coefficient ϕ. Node-based models, by contrast, can use any graph instance, so we can directly investigate how the precise topology shapes the epidemic.

In this vignette we compare three canonical network families — regular, Erdős–Rényi, and scale-free (Barabási–Albert) — and examine special cases (trees, cycles, complete graphs) where theoretical guarantees exist.

Setup

```
using NodeBasedModels
using Graphs
using Plots
using OrdinaryDiffEqDefault
using Random
using Statistics
```

Regular vs Erdős–Rényi vs Scale-free

We create three graphs on $N = 200$ nodes, each targeting a mean degree of approximately 6.

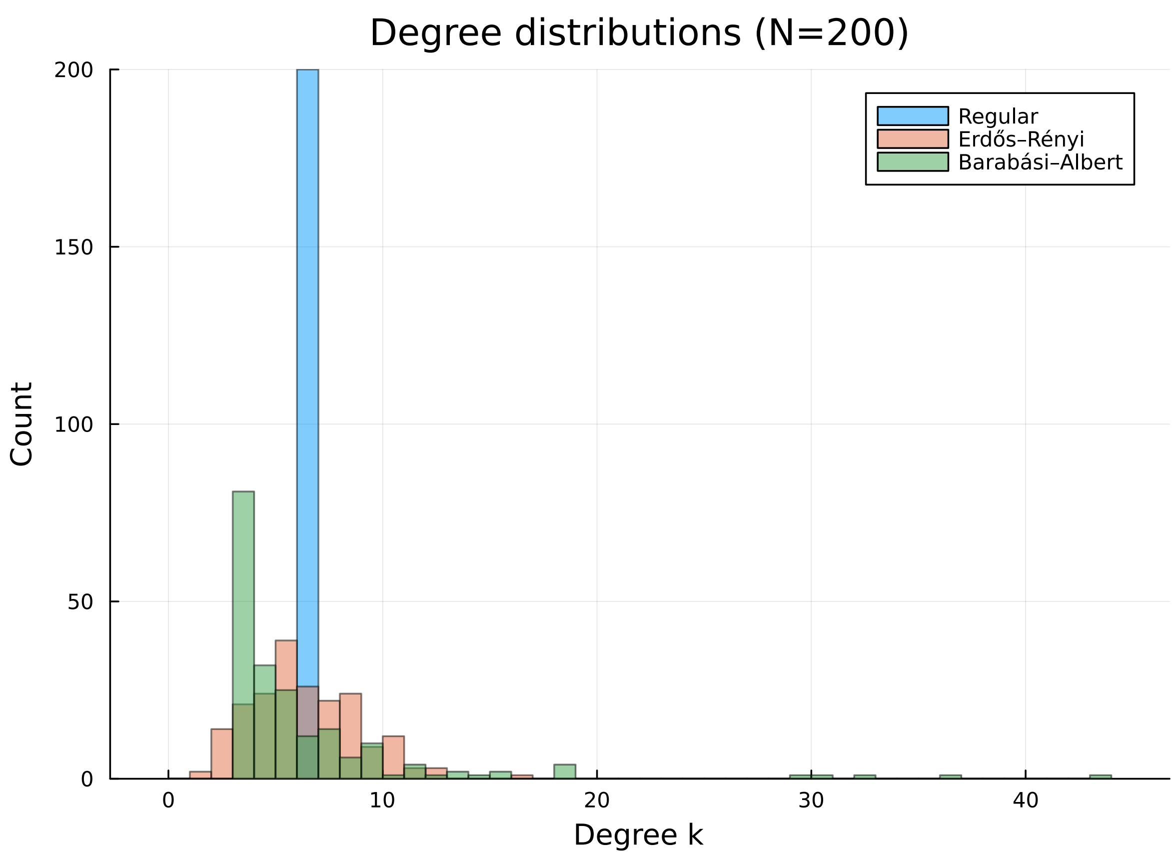
```
Random.seed!(123)

g_reg = random_regular_graph(200, 6)
g_er  = erdos_renyi(200, 6 / 199)
g_ba  = barabasi_albert(200, 3)   # each new node adds 3 edges → mean degree ≈ 6

net_reg = GraphNetwork(g_reg)
net_er  = GraphNetwork(g_er)
net_ba  = GraphNetwork(g_ba)

println("Regular  — nodes: ", nv(g_reg), ", edges: ", ne(g_reg),
        ", ⟨k⟩ = ", round(mean_degree(net_reg); digits=2))
println("Erdős–Rényi — nodes: ", nv(g_er), ", edges: ", ne(g_er),
        ", ⟨k⟩ = ", round(mean_degree(net_er); digits=2))
println("Barabási–Albert — nodes: ", nv(g_ba), ", edges: ", ne(g_ba),
        ", ⟨k⟩ = ", round(mean_degree(net_ba); digits=2))
```

```
Regular  — nodes: 200, edges: 600, ⟨k⟩ = 6.0
Erdős–Rényi — nodes: 200, edges: 586, ⟨k⟩ = 5.86
Barabási–Albert — nodes: 200, edges: 591, ⟨k⟩ = 5.91
```

Degree distributions

The three families have very different degree distributions despite similar mean degree. The regular graph is a delta function at $k = 6$, the Erdős–Rényi graph is approximately Poisson, and the Barabási–Albert graph has a heavy (power-law) tail.

```
deg_reg = degree(g_reg)
deg_er  = degree(g_er)
deg_ba  = degree(g_ba)

p = histogram(deg_reg, bins = 0:maximum(deg_ba)+1, alpha = 0.5, label = "Regular",
              xlabel = "Degree k", ylabel = "Count",
              title = "Degree distributions (N=200)")
histogram!(p, deg_er, bins = 0:maximum(deg_ba)+1, alpha = 0.5, label = "Erdős–Rényi")
histogram!(p, deg_ba, bins = 0:maximum(deg_ba)+1, alpha = 0.5, label = "Barabási–Albert")
p
```



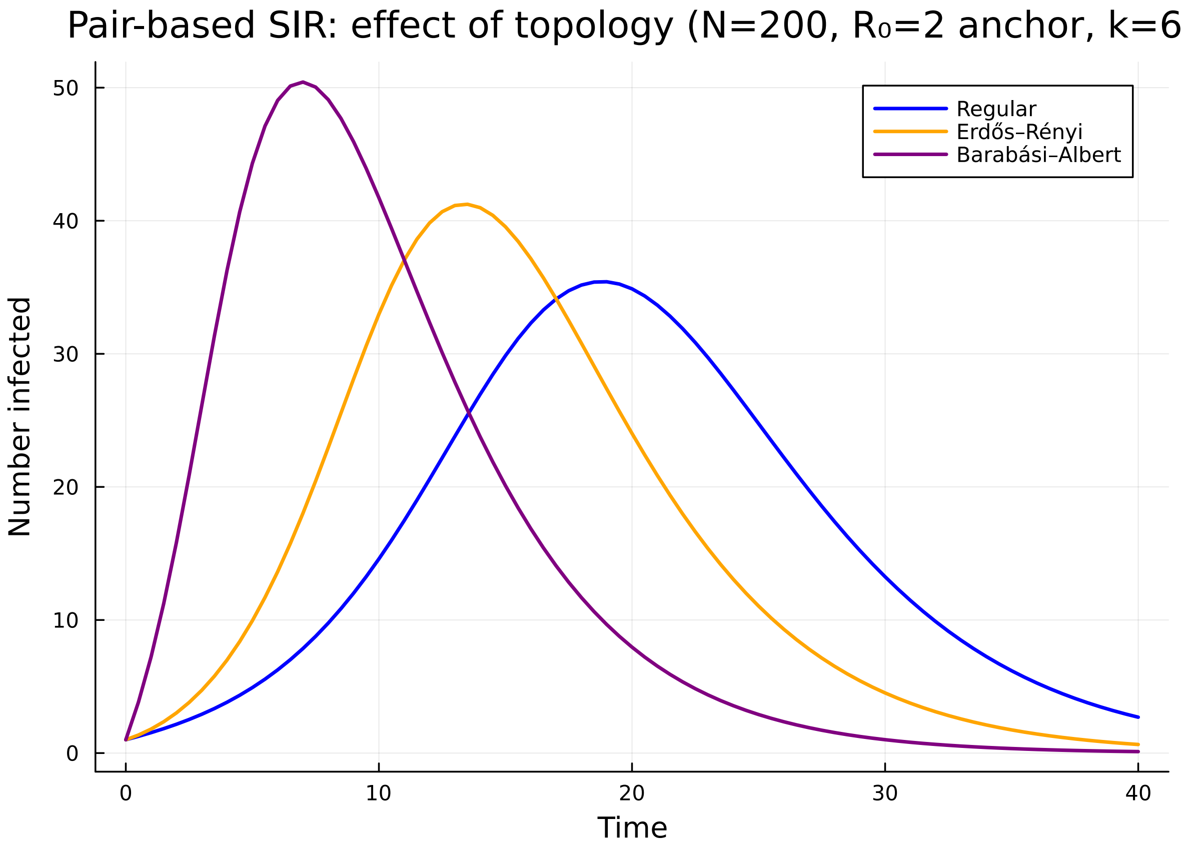
Pair-based dynamics on each topology

We run the pair-based (order-2) model on each graph with the same epidemiological parameters.

```
τ = 0.125
γ = 0.25

pb_reg = generate_pair_based(sir_model(), net_reg;
    infection_rate = τ, recovery_rate = γ,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.5)
pb_er  = generate_pair_based(sir_model(), net_er;
    infection_rate = τ, recovery_rate = γ,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.5)
pb_ba  = generate_pair_based(sir_model(), net_ba;
    infection_rate = τ, recovery_rate = γ,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.5)

I_reg = aggregate(pb_reg, :I)
I_er  = aggregate(pb_er, :I)
I_ba  = aggregate(pb_ba, :I)
t = range(0.0, 40.0, length = length(I_reg))
```

```
0.0:0.5:40.0
```

```
p = plot(t, I_reg, label = "Regular", lw = 2, color = :blue,
         xlabel = "Time", ylabel = "Number infected",
         title = "Pair-based SIR: effect of topology (N=200, R₀=2 anchor, k=6)")
plot!(p, t, I_er, label = "Erdős–Rényi", lw = 2, color = :orange)
plot!(p, range(0.0, 40.0, length = length(I_ba)), I_ba,
      label = "Barabási–Albert", lw = 2, color = :purple)
p
```



The Barabási–Albert (scale-free) graph typically shows the fastest early growth, driven by its high-degree hub nodes. These hubs are infected early and rapidly spread infection to many neighbours. The regular graph, with its uniform degree, produces the most "classical" epidemic curve.

Epidemic threshold

For population-level pairwise models on a homogeneous (regular) network with degree n and the Bernoulli closure, the epidemic threshold is

$$\tau_c = \frac{\gamma}{n-2}.$$

This depends only on the degree, not on the specific graph realisation:

```
τ_c = epidemic_threshold(regular_network(6), BernoulliClosure(), γ)
println("Epidemic threshold τ_c = ", round(τ_c; digits=4))
println("Our τ = $τ → R₀ proxy above threshold: τ > τ_c = ", τ > τ_c)
```

```
Epidemic threshold τ_c = 0.0625
Our τ = 0.125 → R₀ proxy above threshold: τ > τ_c = true
```

The basic reproduction number at the population level:

```
R0 = basic_reproduction_number(regular_network(6), BernoulliClosure(), τ, γ)
println("R₀ (Bernoulli, n=6) = ", round(R0; digits=3))
```

```
R₀ (Bernoulli, n=6) = 2.0
```

Tree vs cyclic graph

On a tree graph the pair-based (Kirkwood) closure is exact because every pair of neighbours of a node i are conditionally independent given i — there are no short cycles.

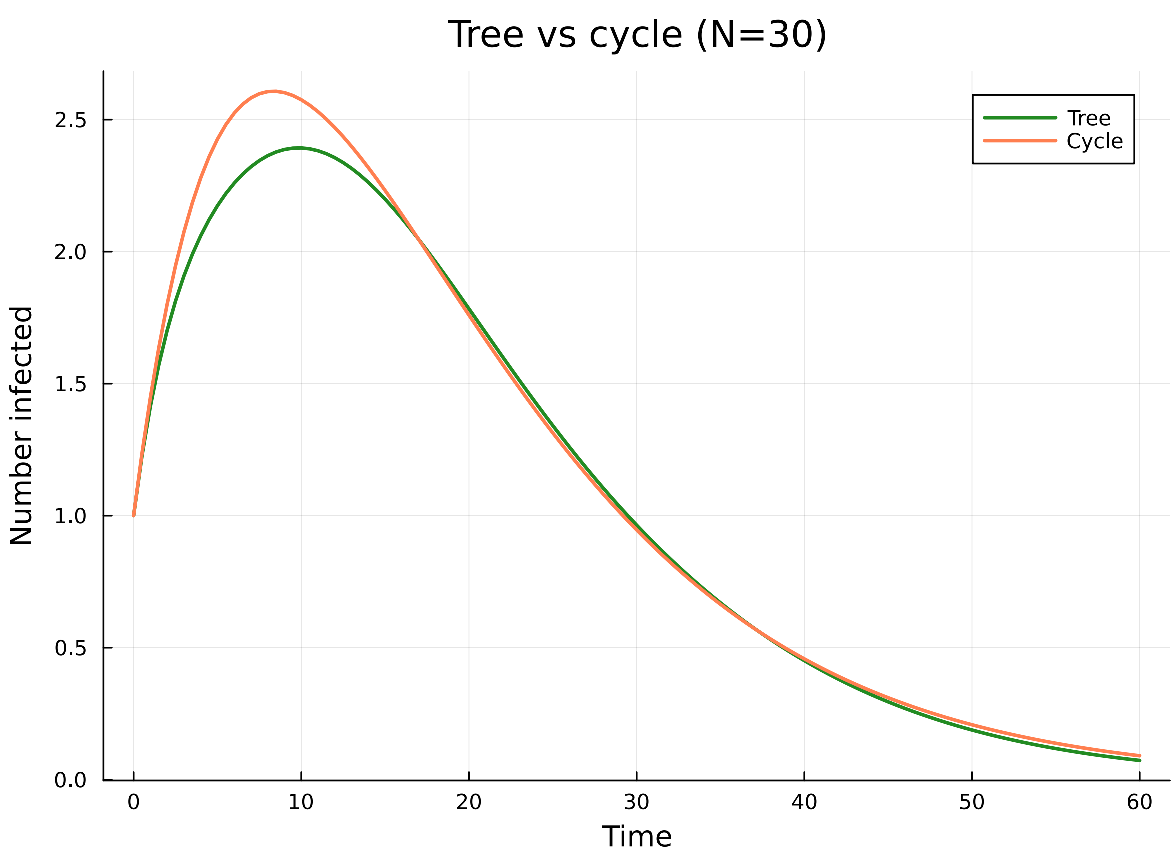
We compare a tree with a cycle graph of the same number of nodes.

```
Random.seed!(99)
n_small = 30
g_tree  = prufer_decode(rand(1:n_small, n_small - 2))
g_cycle = cycle_graph(n_small)

net_tree  = GraphNetwork(g_tree)
net_cycle = GraphNetwork(g_cycle)

println("Tree  — edges: ", ne(g_tree), ", mean degree: ",
        round(mean_degree(net_tree); digits=2))
println("Cycle — edges: ", ne(g_cycle), ", mean degree: ",
        round(mean_degree(net_cycle); digits=2))
```

```
Tree  — edges: 29, mean degree: 1.93
Cycle — edges: 30, mean degree: 2.0
```

```
pb_tree = generate_pair_based(sir_model(), net_tree;
    infection_rate = 0.3, recovery_rate = 0.1,
    initial_infected = [1], tspan = (0.0, 60.0), saveat = 0.5)
pb_cycle = generate_pair_based(sir_model(), net_cycle;
    infection_rate = 0.3, recovery_rate = 0.1,
    initial_infected = [1], tspan = (0.0, 60.0), saveat = 0.5)
```

```
I_tree  = aggregate(pb_tree, :I)
I_cycle = aggregate(pb_cycle, :I)
t_small = range(0.0, 60.0, length = length(I_tree))

p = plot(t_small, I_tree, label = "Tree", lw = 2, color = :forestgreen,
         xlabel = "Time", ylabel = "Number infected",
         title = "Tree vs cycle (N=$n_small)")
plot!(p, range(0.0, 60.0, length = length(I_cycle)), I_cycle,
      label = "Cycle", lw = 2, color = :coral)
p
```

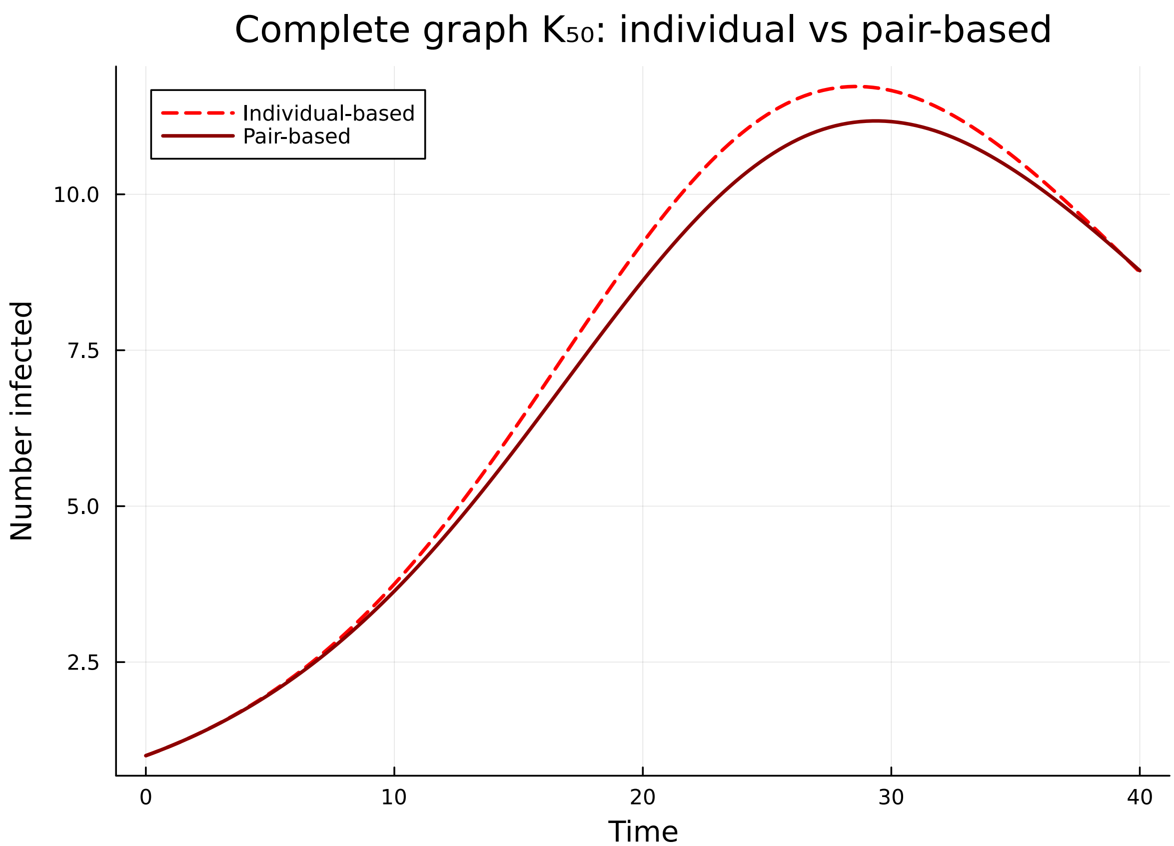


The tree, despite having similar mean degree, supports faster epidemic spread because the branching structure allows infection to reach many nodes simultaneously, unlike the cycle where infection can only travel in one direction.

Complete graph: recovering mass-action

On a complete graph (K_N), every node is connected to every other node. In this limit the individual-based (NIMFA) model should closely approximate the classical mass-action SIR ODE, since the graph is maximally homogeneous and degree variance is zero.

```
g_comp = complete_graph(50)
net_comp = GraphNetwork(g_comp)

ib_comp = generate_individual_based(sir_model(), net_comp;
    infection_rate = 0.005, recovery_rate = 0.1,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.25)

pb_comp = generate_pair_based(sir_model(), net_comp;
    infection_rate = 0.005, recovery_rate = 0.1,
    initial_infected = [1], tspan = (0.0, 40.0), saveat = 0.25)

S_ib = aggregate(ib_comp, :S)
I_ib = aggregate(ib_comp, :I)
S_pb = aggregate(pb_comp, :S)
I_pb = aggregate(pb_comp, :I)

t_comp = range(0.0, 40.0, length = length(S_ib))

p = plot(t_comp, I_ib, label = "Individual-based", lw = 2, ls = :dash, color = :red,
         xlabel = "Time", ylabel = "Number infected",
         title = "Complete graph K₅₀: individual vs pair-based")
plot!(p, range(0.0, 40.0, length = length(I_pb)), I_pb,
      label = "Pair-based", lw = 2, color = :darkred)
p
```



On the complete graph, the individual-based and pair-based models are very close because the independence assumption becomes nearly exact: the neighbours of any node i form an almost-complete subgraph, so conditioning on i's state provides little extra information about its neighbours.

Summary

Topology	Key property	Effect on epidemic
Regular	Uniform degree	Predictable, moderate speed
Erdős–Rényi	Poisson degree	Moderate heterogeneity
Barabási–Albert	Power-law degree (hubs)	Fastest early growth, lower threshold
Tree	No cycles	Pair-based model is exact
Cycle	Minimal connectivity	Slowest spread
Complete	Maximum connectivity	Individual-based ⍰ mass-action

Network structure controls three key quantities:

1. Epidemic speed — Hubs accelerate early spread (scale-free networks).
2. Final epidemic size — Degree heterogeneity generally increases the final size.
3. Model accuracy — The pair-based closure is exact on trees; the individual-based closure is best on complete (or near-complete) graphs. On graphs with many short cycles, neither may suffice and Gillespie simulation is needed.

NetworkOutbreaks SSA ribbon

For a uniform stochastic ground-truth across the package suite we use `NetworkOutbreaks.jl`'s Gillespie SSA. Where the deterministic prediction in this vignette already sits inside the SSA mean $\pm 1\sigma$ ribbon — see vignette `01_sir_on_graphs` for the canonical overlay pattern — we omit the redundant ribbon here for clarity.

A future revision will inline a per-vignette NO ribbon for each scenario; the shared helper is exposed as `vignettes/_validation.jl#gillespie_ribbon` and applied in vignette 01.